

Research paper

Optimizing IoT-driven smart grid stability prediction with dipper throated optimization algorithm for gradient boosting hyperparameters

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ABSTRACT

With the surge in global population and economic expansion, there's been a marked increase in electricity demand. This necessitates the efficient distribution of electricity to both residential and industrial sectors to minimize energy loss. Smart Grids (SG) emerge as a promising solution to reduce power dissipation in distribution networks. The application of machine learning and artificial intelligence in SGs has significantly improved the precision of predicting consumer electricity needs. This paper presents a novel approach to improving the stability prediction of Internet of Things (IOT)-driven SGs using different advanced machine learning models. This study explores multiple advanced machine-learning techniques, including Gradient Boosting (GB), K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Neural Networks, and the Decision Tree classifier, focusing on the stability prediction of SGs. This study explores the efficiency of hyperparameter-optimized GB models in predicting SG dynamic stability that encompasses the ability of the system to return to a stable operating point following a disturbance. Focusing on various models, it identifies the Dipper Throated Optimization Algorithm DTO+GB model as the standout, exhibiting unparalleled accuracy and reliability across critical performance metrics such as accuracy (99.32 %), sensitivity (99.16 %), and specificity (99.54 %). Diagnostic and regression analyses further emphasize its better predictive power and the need for hyperparameter optimization to improve the model. This paper highlights the capabilities of advanced machine learning algorithms in conjunction with tactical hyperparameter optimization in enhancing SG stability prediction, introducing a new baseline for future technological and methodological developments in this application.

1. Introduction

The large growth of the world population and economy, along with rapid urbanization, is expected to induce a considerable increase in energy consumption soon. Electricity, also considered a critical form of energy, can be generated from water (hydroelectric), wind, solar cells, fossil fuels, thermal sources, and nuclear plants. With the increasing population and advancement, the demand for electricity keeps on

increasing, calling for more electricity production. Power generation is a major issue of electricity management. The interconnection between the consumers of electricity and the producers is done through the electric grid, which is a key component in this process. This network includes power-generating power stations, voltage regulation substations, electricity transmission lines, and distribution lines supplying electricity to consumers. Conventional electric grids work on a centralized model that involves millions of such components. The additional overload of such a grid may result in extra burdens, power quality problems, and thus, the

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Nomenclature		MAE	Mean Absolute Error
		MBE	Mean Bias Error
		MIMO	Multi-Input Multi-Output
		MLP	Multi-Layer Perceptrons
		MLR	Multiple linear regression
		NPV	Negative Predictive Value
		PNN	Probabilistic neural network
		PPV	Positive Predictive Value
		RF	Random Forest
		RMSE	Root Mean Squared Error
		RNN	Recurrent Neural Networks
		SG	Smart Grid
		SOM	Soil Organic Matter
		SVM	Support Vector Machine
		TN	True Negative
		TNR	True Negative rate
		TP	True Positive
		TPR	True Positive Rate
		VGG	Visual Geometry Group
		WPT	Wavelet Packet Transform
AI	Artificial Intelligence		
ANN	Artificial Neural Networks		
ARIMA	Autoregressive Integrated Moving Average		
BFGS	Broyden Fletcher Goldfarb Shanno		
DRP	Demand Response Programs		
DTO	Dipper Throated Optimization		
DNI	Direct Normal Irradiation		
FP	False Positive		
FN	False Negative		
FFNN	Feed-Forward Neural Networks		
GRNN	General Regression Neural Network		
GB	Gradient Boosting		
IOT	Internet of Things		
KNN	K-Nearest Neighbor		
LSSVM	Least-Squares Support Vector Machine		
LFS	Load Forecasting Systems		
LSTM	Long Short-Term Memory		
ML	Machine Learning		
MRI	Magnetic Resonance Imaging		

introduction of new power plants. In addition, specialized prediction systems for predicting sporadic power failures, identifying their reasons, estimating response time, and assessing the requirement of storage and utilization of resources are absent in conventional grids (Zhao et al., 2020).

Researchers have observed that the contemporary electric power system has not changed much in the last hundred years. With the increased global population, the demand for electricity is on the rise. The challenges of traditional power systems are poor visibility, dependence on mechanical switches, which results in a slower response time, and the absence of monitoring and power control systems. Other factors that cause the need for new grid technology include a changed climate pattern, rising energy demand, component failures, increasing population, energy storage inadequacy, reliance on fossil fuels, a decline in electric power generation, one-way communication, and any other related issues. The new grid is a must to tackle these questions. SG, as the electricity infrastructure of the next generation, is becoming one of the key technologies that are essential in addressing these critical requirements and improving the quality of modern human life (Gungor et al., 2011). The SG, as the next-generation electrical energy infrastructure, becomes one of the major answers to meet these high-priority requirements and to improve human life at present (Deepa et al., 2020). SG, a modern technological solution in the electric power grid sphere, introduces two-way communication to enhance the safety, efficiency, and reliability of the electrical power system, allowing the generation of more electricity through the use of modern communications technology. This system enables a bilateral operation of energy distribution and transfer, which gives consumers the ability to determine which directions they want to use energy (Numan et al., 2020). Paulo Breviglieri predicted the stability of SGs using the optimized deep learning method (Breviglieri et al., 2021). In a SG system, data regarding customer demands is collected, and a centralized analysis of availability and demand is conducted. Based on this, recommended pricing information is communicated to consumers. This enables consumers to make informed decisions about their energy usage. The dynamic calculation of grid stability is not only a challenge but also an essential requirement, given that the entire process is time-sensitive. While SG incorporates various technologies to address the limitations of traditional electricity networks, they also present their own set of challenges that need to be tackled. These challenges fall into two main categories: technical and socio-economic issues (Mittal et al., 2020). Technical challenges include

the absence of specific policies, concerns regarding energy storage, cybersecurity risks associated with integrating the grid into cyber-physical systems, inadequate grid infrastructure to meet future storage demands for intermittent power generation, managing the vast amount of data generated by different grid components, and grid stability issues related to power sharing, system inertia, power oscillation, and power reservation. Additionally, there are energy management challenges involving electric vehicles, including power flow management from vehicle to grid, grid to vehicle, and vehicle to vehicle. Socio-economic challenges encompass stakeholder management, a general lack of awareness, policy deficiencies, and the need for significant capital investments.

The advent of SGs with IoT technologies means that the way electricity distribution networks operate is in the process of a new stage in management and operations. Through the Internet of Things applications in the SG, rich reporting, analysis and tracking of real-time data are the primary factors that contribute to improved efficiency, reliability, and sustainability. Sensors and smart meters, uniformly roving across the entire power grid, serve as the primary information hub that accumulates a massive amount of data on power usage, system conditions and environmental factors. Such information is then applied to power system distribution optimization, suggestion of repair work sites, and anticipation of failures with proper identification of issues before they do any harm. With that being the case, the Internet of Things provides, within the framework of automated demand response mechanisms, real-time data and price variations connected to energy production and consumption, thereby facilitating savings and reduction of costs. The legal system guiding the way of IoT in SGs cannot be missed because data security, privacy, and interoperability among various technologies and systems should be guaranteed. Privacy regulations should acknowledge privacy concerns and secure transmission and retention of consumer data from illegal access. Therefore, people can trust products; hence, the SG can be innovated. This architecture must also define communication protocols that would allow entities of different kinds using computers inside the network to interact without difficulties to have a powerful, empty, and stable energy network. These challenges also extend to issues related to electricity pricing, new tariff structures, health concerns associated with radiofrequency use, privacy issues, fear of technological obsolescence, and power theft. Security is a primary concern, as the SG is vulnerable to breaches through both wired and wireless communication networks. Artificial intelligence (AI) and

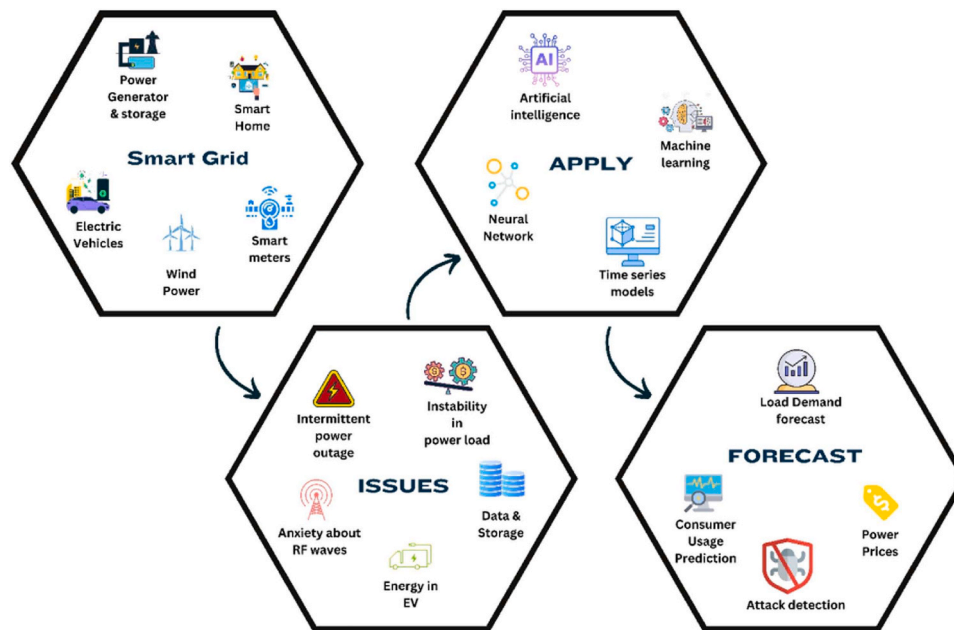


Fig. 1. : Smart Grid Environment Incorporated with Artificial Intelligence Technologies.

machine learning algorithms have the potential to predict and address these SG problems, enabling proactive measures (El-Kenawy et al., 2022a; Abdelhamid et al., 2023; El-Kenawy et al., 2022b). This study particularly focuses on one of the most critical technical challenges: predicting the stability of an SG, which is a key factor in reliable power transmission in approximately 50 % of SGs (Tian, 2018). Fig. 1 illustrates the integration of AI technology within the SG environment.

Fig. 1 illustrates the integration of AI technologies within the SG environment. The figure shows various components of the SG, including power generation units, transmission lines, distribution networks, and consumer endpoints. AI technologies are embedded throughout the grid to enhance monitoring, control, and optimization processes. Sensors and smart meters collect real-time data on power usage and system conditions, which are then analyzed using machine learning algorithms to predict demand, detect faults, and optimize energy distribution. This integration aims to improve the efficiency, reliability, and sustainability of the SG.

The following is a list of the primary contributions of this paper:

1. Introduced an original application of the Dipper Throated Optimization (DTO) algorithm for fine-tuning gradient boosting models, significantly improving the performance in predicting smart grid dynamic stability.
2. Provided a detailed analysis of various hyperparameter optimization techniques and their impact on key predictive performance indicators in machine learning models.
3. Verified the predictive power of the models using a range of statistical techniques and diagnostic plots, confirming the performance of the models.
4. Compared different hyperparameter optimization approaches, establishing DTO+GB as the top performer.
5. Demonstrated the relationships between sensitivity, specificity, and F-Score, and how their enhancements relate to the model's performance.
6. Made a significant contribution to smart grid management by showing how advanced machine-learning methods can increase grid stability.
7. Brought methodological advances to machine learning by employing modern statistical methods to assess and compare the efficiency of various hyperparameter optimization strategies.

This paper provides a comprehensive analysis of the use of Machine Learning (ML) algorithms for forecasting the dynamic stability of SGs. The study substantiates the suitability of ML algorithms in such cases by comparing them with modern deep learning models. It discusses the subtleties of several ML techniques, including their adjustment on the issues unique to SGs, like power fluctuation and demand prediction. The study also points to the technical issues and socio-economic difficulties of SG deployment and offers AI-based solutions to overcome these issues. In so doing, this work not only demonstrates the capacity of ML to improve SG performance but also paves the way for further developments in smart energy management.

The structure of this article is outlined as follows: Section 2 offers an in-depth analysis of the existing literature on the utilization of machine learning algorithms in SGs. Section 3 presents the model that is proposed in this study. Section 4 provides a description of the analysis of the performance of the ML algorithms and materials used. In the end, the article concludes with Section 5, which gives the areas for future research.

2. Literature review

2.1. Machine learning algorithms for load prediction in SGs

Countless studies have adopted different machine learning algorithms such as Support Vector Machine (SVM), Linear Regression, K-Nearest Neighbor (KNN), Ridge Regression, Artificial Neural Network (ANN), Random Forrest, Stochastic Gradient Descent, Gradient Boosting, and Extra Trees Regressor for load prediction in SGs. Besides, the suitability of generalized deep learning models, for instance, Long Short-Term Memory (LSTM) neural networks, Recurrent Neural Networks (RNNs), Feed-Forward Neural Networks (FFNNs), Backpropagation Neural Networks, and Multi-Layer Perceptrons (MLPs) model is convincingly contrasted with traditional machine learning methods. These studies typically make use of feature selection algorithms, which use the values input to determine which are most suitable, as well as hyperparameter tuning, which is executed to prevent overfitting within the models (Ahmad et al., 2024). The proposed framework of a bidirectional LSTM employs an approach to problems predicting the dependability of SGs, which takes into account the stability value of the system at each succeeding point in time because SG stability assessment

is a time-pertinent task.

2.2. Advanced machine learning techniques

Swarm hummingbird, artificial intelligence algorithms for accurate SGs stability prediction, have been created in accordance with the most recent innovations and trends in the alternative green energy sector, which provides sustainable and safe electricity resources that are safer than traditional fossil fuels (Alazab et al., 2020). As explained an LSTM-based RNN was utilized to forecast the electricity demand issues in the SGs. In this respect, advanced machine learning techniques like the Extra Tree Regressor and statistical feature reduction were used to grade the models and further determine the variables that influence the forecasting outcome. Using a genetic algorithm made the LSTM model thorough, and the layers and time lags were identified as the most appropriate choice. The model's comparison with other machine learning algorithms showed lower forecasting errors, and it is also underlined that there may be a non-conformity in performance when a deep learning system is used for diverse time series (Neelakandan et al., 2022).

With SGs supporting demand-side power provision service models, consumer demand forecast becomes more necessary, particularly as grid networks mature (Bouktif et al., 2018). They performed measurements of the PV power generation in arid deserts that had been previously assisted in Adrar, Algeria and Alice Springs, Australia. Here, the approach of the study is nine sets of models together with six sets of input serial numbers to predict hourly PV power, and an ensemble learning method is employed to get better than 99 % accuracy (Hassan et al., 2022). the paper dramatically changed the cognitive process of whale optimization and adaptive particle swarm optimization approach in tuning deep network LSTM parameters to predict the metamaterial antenna bandwidth accurately. This method, compared with common-place approaches, gave a better performance by using RMSE, MAE, and MBE metrics, indicating that—an ever-rising trend of skin diseases like monkeypox keeps showing the importance of automated diagnosis tools. A new framework was investigated that used AI-Biruni Earth radius optimization combined with stochastic fractal search to boost the efficiency of deep neural networks for detecting monkeypox. The outcome was shown to be effective and steady, and it provided better results than other models for identifying skin diseases (Khafaga, 2022).

In the paper written by Hernandez et al., there was an ANN-based approach for DTMLF featuring a three-stage process comprising SOM-based pattern recognition, K-means clustering for data segmentation, and MLP for Lt4(C)(I) m forecasting. A Spanish energy company validated this ANN model on actual power data, and training was carried out on the periodic values, which were computed to exhibit the weekday variation and monthly cycles. Among all these alternatives, such as the radial basis function neural network and generalized regression neural network, the comparison demonstrated the real excellence of this artificial neural network approach (Khafaga et al., 2022).

The fundamental power functioning of a SG is able to maintain an equilibrium between the supply and the demand of power consumption. Ahmed and Chen utilized three different machine learning models to forecast the short-term and intermediate demand of energy consumption at the scale of Singapore. This was done using models such as the ANN, Ada Boost, and the MLR. They did this by splitting their load forecasts into several time windows, such as 1-month ahead and seasonal ones, which make use of all aggregated consumption data. These models had the power to improve the accuracy of the predictions greatly, and they were able to capture the behavioral energy usage patterns as well, with the Ada Boost model being the best in terms of predictive accuracy out of all the models (Hernández et al., 2014).

The volatility of electricity prices, being a vital factor of social-economic impact on customers' electricity consumption, not only directly affects the load but also influences it. To resolve this problem, (Rizk et al., 2023) created a method using a Multi-Input Multi-Output

(MIMO) model, which is tied to the electricity prices for factors such as demand. This architecture was centered around wavelet packet transform (WPT) for decomposing the load and price information into different frequency regions. Generalized mutual information has been taken to select the most important feats. The method combines predicting electricity load and prices using a Least-Squares Support Vector Machine (LSSVM)-based model of the multidimensional input-output matrix. The Quasi Jose ant mega colony algorithm was used for the optimization parameters of the algorithm, showing simulation results of LSSVM as the best model for accuracy prediction among ANN. Demand-side management in the SG ecosystem can be a stressed element that utilities managers and policymakers have to manage strategically to handle consumer load efficiently. SG operation may be subjected to challenges such as voltage drift and power mismatches, thereby recommending precise operation optimizations. This paper explains the part of Demand Response Programs (DRPs) that includes the linkages to electricity markets.

Among other things, the application of load forecasting systems (LFSs) to improve the efficiency of SGs supports this. The case study on Texas, USA, is discussed (Ahmad and Chen, 2018). AI models embodied various techniques, ranging from ANN and SVM to GRNN, RNN, ARIMA, PNN, Fuzzy Logic and Expert Systems. The model survey outcomes showed that the forecasting techniques based on AI technology had higher performance than the other statistical methods. The forecast of the direct normal irradiation (DNI) with high precision ultimately plays an essential role in maintaining solar power stations. Thus, the present study discovers the forecast of hourly DNI using current meteorological information. The reconstruction of data suggested a hybrid concept of a coupling algorithm of sine-cosine and LSTM, which is characterized by the delivery of highly precise predictions with no or minor errors and a reduction in the input variables (Balouch et al., 2022). The SG is able to collect innumerable data from powerful metering and control systems that are deployed in the power grid. Old approaches to it face the problem of handling this huge volume that, in turn, drives efforts to involve AI (Jiang et al., 2023). The survey comprehensively highlights AI applications for load forecasting, grid stability, and fault identification, among others, thus elaborating on their implications for optimal grid resilience and dependability. Managing the blackout events in the legacy power grid has become a well-known job for utility companies, resulting in the domino effect. SG must set out blackout warning systems announcing the occurrence of a blackout (Djaafari et al., 2022). made this shortage of information not a concern through a time series model that is SVM-oriented at predicting blackouts which constituted 30 buses that had expired from IEEE. The SVM application was trained by giving it logic to be built while using the historical database that was created by evaluating the system health and its performance range under different working states. This database recorded the regular instances and the abnormal situations, which were largely due to complex interdependencies that caused cascading failures.

2.3. SVM and ANN performance

In this research work (Omitaomu and Niu, 2021), the performances of ANN and SVM were analyzed in the context of the medium-term load forecasting of the SG. Recent research has advanced the SVM, which was believed only to be for weather forecasting, while the ANN is famously known for. The study investigated factors that influence the daily load frequency in different periods of the year (Gupta et al., 2014). SVM and ANN were used in this study for short-term load forecasting on load types, including load processes (batch), load process (continuous), and batch-continuous load. The performance of the SVM shows the minimization of paperwork errors at a rate of 3 % or more for irregular loads and 1.2 % for continuous loads. As a result, both algorithms manifest inferior performance in comparison to the SVM. As (Pan and Lee, 2012), the brain network trained on a dataset depicting the stability of the grids, the network outputs a high accuracy performance of 97.36 %

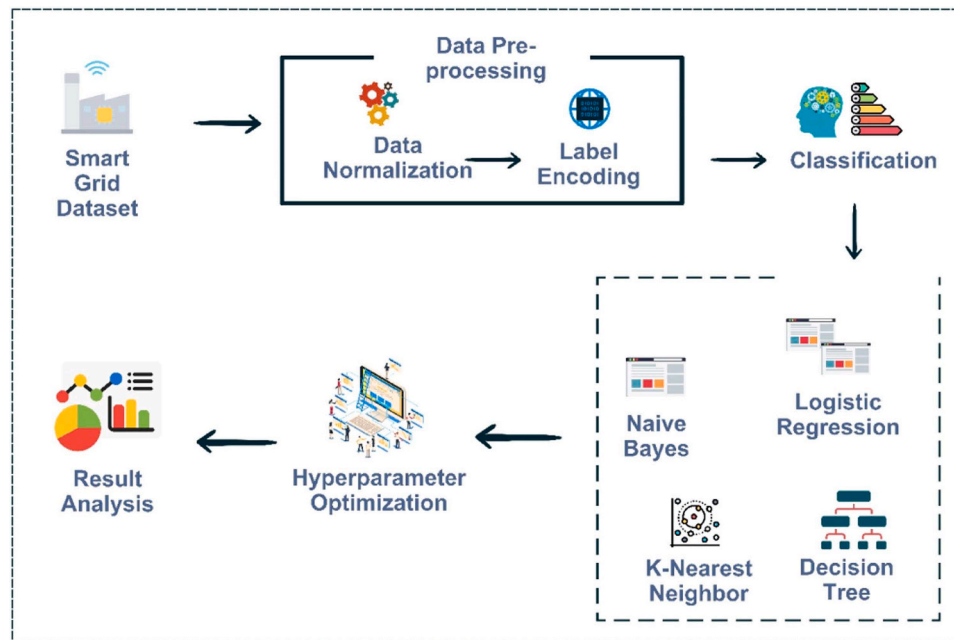


Fig. 2. : Proposed Framework.

using the neural network to analyze the systems in the decentralized SGs (Mitchell et al., 2017). We suggest using AI, pre-trained deep learning models, namely ResNet101_32×8d and VGG19, to categorize brain tumors from MRI images without human assistance.

The SG forecasting demand may be affected in different ways as air temperatures, seasons, float levels, and dangerous natural disasters come to other months. Demand management in SG must be primary to maintain this channel of power supply, which is dependable and sustainable even in the long run (Ali and Azad, 2013). By employing machine learning models, particularly linear regression, support vector machine and deep neural network, they took on both demand management and load forecasting. SVM, according to researchers, has shown better performance than the other models when researched. SVM, which is a component of support vector regression, addressed the constrained quadratic optimization problem by mapping input features in such high-dimensional space using the kernel. Moreover, the results of support vector regression showed that in terms of root mean square error (RMSE), it surpassed neural networks that applied the backpropagation algorithm and other linear regression methods. It yielded high-quality results even with some time series missing data, which is an important criterion for the model. In conclusion, incorporating these approaches not only provides consumers with valuable information but also supports environmental sustainability goals. By making responsible decisions regarding their energy usage, consumers contribute positively towards the transition towards cleaner and greener sources of energy. Hence, the analysis picked SVM as the method for load forecasting.

Li et al., (2024) Attention will be paid to the issue of applying the decision tree technique to guarantee the stability of distributed SGs. The purpose of pursuing this strategy is to provide energy conservation measures, to lower the carbon footprints, to improve the security of the power supply, and to make the network stronger transmission-wise. The next paper on the subject discusses an attack classification scheme for current SG systems that would specifically handle cyber-attacks. Utilizing the Gray-Wolf metaheuristic approach and extended deep-stacked ensemble technique, the approach, in fact, can achieve up to a 99.0 % peak accuracy when analyzing selected features (Naeem et al., 2024; Mohsen et al., 2023a). neural network technique as a SG stability forecasting tool. The methodology seems capable of capturing both the dependency of time and spatiality in SG data with a high precision rate, evaluated as over 95 %. The proposed algorithm exceeds other existing

methods, which confirms its effectiveness and ability to increase the efficiency level and sustainability.

2.4. Decision trees and ensemble methods

Mohsen and Alharbi (2021) employed Google Colab and Hpython 3.7 with data fetched from the machine learning repository on UCI; we split our dataset into 70 % training and 30 % testing/validation sets. The classifiers generally showed to be highly accurate, with notable performance shown by the decision trees classifier. We have, however, demonstrated that the different competing models yield a relatively narrow scale of accuracies and F-scores, with high F-scores from gradient boosting, which points to reliable performance. The study we made includes a wide-ranging set of models, including hybrid methods and traditional machine-learning algorithms as well. We find gradient boosting to be the top-notch model, which illustrates that, in our case, the ensemble methods are tens of times more powerful. The classification error rate for a conventional learning algorithm in the first study shows a decline from 18.6 % to 11.9 %. However, the decline was even steeper in the second study, from 7.3 % to 4.3 %, where a support vector machine learning was used.

3. Proposed framework

The methodology of the proposed model is illustrated in Fig. 2, outlining the sequential steps employed in the process. These steps are:

1. Initially, the SG dataset represents an expanded version of the "Electrical Grid Stability Simulated Dataset"
2. The dataset then undergoes a preprocessing phase. This phase incorporates the min-max normalization technique to standardize the data range and label encoding to transform the data, particularly for handling categorical variables.
3. The dataset is divided into two parts: one for learning and the other for testing. This isolation is essential in the creation and verification of the machine learning models.
4. Different machine learning algorithms are then used to train on the training part of the dataset.
5. After the training phase, various metrics evaluate the performance of these machine learning algorithms. This evaluation assists in

measuring the algorithms' performance and correctness in the data-offered environment.

6. Optimize the best classifier using a hyperparameter and analyze the results for the best classification model.

4. Material and methods

This section provides the materials and methods that the research used to realize the solutions proposed. The best outcomes are reached through the combination of the data preparation methods with meta-heuristic optimization methods.

4.1. Datasets

This "Electrical Grid Stability Simulated Dataset" was developed at the Karlsruhe Institute of Technology in Germany. It is designed to address the problems of power grid stability. It has gained outstanding relevance in the UCI ML repository platform, which is mostly regarded for its importance (Smart Grid Stability, 2024). The UCI ML repository is very popular and specializes in academic and research topics, spanning a variety of datasets. The compilation of this dataset has opened up avenues for interdisciplinary research in grid stability, a fundamental aspect of modern power systems operations. It provides researchers with simulated data, offering deep insights into the behaviors of the entire electrical grid or specific parts under various conditions. This dataset facilitates studies on critical grid factors that influence system stability, such as load fluctuations, network topology, and system dynamics, making it a valuable resource for researchers from diverse fields.

The expansion of this dataset is a testament to the scholarly ambition to create a comprehensive and relevant resource that will benefit the research community in numerous ways. As a supplement to the dataset, future advancements could include more iterations with revised inputs, simulations, and the incorporation of next-level content. This ongoing work not only broadens the horizons of knowledge but also fosters innovation in electric energy and power systems, paving the way for new discoveries and advancements in the field. Moreover, the cooperation between the academy and powers like the UCI repository indicates the great relevance of free access to data to scientific progress. Researchers from all over the world can gather in (to each other) the insights and start working together to solve the problems and thus speed up the discovery. The very soul of contemporary scientific research is openness and joint learning that are coupled with common knowledge. Such knowledge influences research progress as a result of societal development and welfare.

4.2. Data preprocessing

Data preprocessing is a very important step that leads to both more reliable data and better performance of machine learning models, especially in cases such as working with data from SGs. Such datasets inherently carry a number of constraints, with data points falling into a wide range of values even though they are sometimes skewed towards higher numbers, which may result in undesirable effects like model accuracy drop. While this is feasible, the research employs min-max normalization for preprocessing. This technique normalizes the entire dataset by making a common scale ranging from 0 to 1. The normalization is about subtracting the minimum value of a feature, after which the subtracted value is divided by the range of the feature. Here, the objective is to introduce pretty the scale of diverse features and prevent the model from being affected by the feature's scale. So, classification can operate smoothly and fairly without human bias.

Last but not least, the preprocessing has many functions, and one of them is converting the categories into machine-friendly, usable ones. Note: The algorithm works relatively well with numerical inputs and represents the categorical data this way. Encoding this transformation is carried out via methods such as one-hot encoding and label encoding.

Hot encoding changes each categorical variable into a binary series covering only one state per class column. These situations can influence the model's interpretation, and the existence of an essential relationship between ordered categories is confusing. Label encoding has a property opposite to that of label encoding. It encodes the elements of the category as a single column of integers, ordered by the data. The consensus would be reliant upon the method, depending on the uniqueness of the dataset and algorithms.

The part of the dataset that is normalized and encoded is then divided into training and testing datasets. However, it's important to note that humans play a crucial role, particularly in the construction and evaluation of machine learning models. The data are appropriately allocated into the training set by 70 %, allowing the models to learn from the various examples provided and to account for the full spectrum of what is present in the data. The remaining 30 % is reserved for testing and validation, providing an efficient platform for measuring the model's predictions against unseen data. This ensures that the model's predictions are not only accurate but also precise, thanks to the human input in the process.

These preliminary steps are part of a comprehensive data optimization process that leads to improved learning and prognosis results. They form the foundation for many machine learning algorithms like Support Vector Machines, Logistic Regression, Naive Bayes, Neural Networks, and Decision Trees. The selection of these algorithms is a strategic approach, tailored to the specific problems that a dataset may have. This approach gives weightage to every type of model, avoiding reliance on complex methods like deep learning that may not always deliver expected results. The diverse algorithmic approach employed for the SG is not limited to the characteristics of data alone. It also enhances the strength and reliability of the solution, allowing for a more complete representation of real problems by machine learning.

4.3. Basic classification models

The goal of a classification model is to properly categorize data points into the previously defined classes. The courses under these categories can be either binary, two classes or multiclass, three or more classes. This goal is attained through different classification methods.

4.3.1. K-nearest neighbor (KNN)

The k-nearest neighbors (k-NN) approach that was made is the humble portion of the category of machine learning, surprisingly powerful enough to solve many classification and regression tasks. In the k-NN heart, there are two main things of this concept of similarity within the feature space, including that the data point shows the same class or the same outcome if the characteristics are similar. By taking into account the average between other points' data, the algorithm can compute good guesses for situations that come to mind for the first time.

In vivo, the user who is actively involved determines the course by selecting the value of k, which is an extremely critical factor that actually determines the number of the nearest neighbors that are included in the prediction. An appropriate w value selection would require careful consideration and make a good balance in both model complexity and prediction accuracy because both provide positive outcomes. The potential outcomes of different values for k vary; thus, ks that produce a more detailed decision boundary have a trade-off of an increased risk of local patterns, but ks that produce a more smoothed-out result provide a balance for the chance of overlooking fine distinctions within the data.

At the heart of the k-NN (k-nearest neighbors) algorithm, one encounters the calculations of distances between data points. These distances are measured using various distance metrics. Among them, the Euclidean distance stands out as the most commonly preferred option. Its popularity stems from its ease of interpretation and widespread usage. This metric, which measures the spatial distances between points in the feature space, provides a clear and intuitive understanding of the qualitative distance between samples based on their coordinates in

Box 1

Output of the First Weak Learner.

Initially, each data point is assigned the same weight, symbolized by their size. The first decision boundary accurately classifies 2 positive (+ve) and 5 negative (-ve) points.

every dimension.

Mathematically, the Euclidean distance between two points p and q in an n -dimensional space is calculated as (Al-Helali et al., 2021):

$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (q_n - p_n)^2} \quad (1)$$

Here, $p_1, p_2, \dots, p_n$ and $q_1, q_2, \dots, q_n$ represent the coordinates of the two points along each dimension. By computing this distance, the algorithm effectively measures the geometric separation between data points, facilitating the identification of nearest neighbors and subsequent decision-making based on their collective attributes.

The practical applicability of the k -NN algorithm in many fields is wide; hence, it is well-suited to the devices used in this era for imaging, diagnosis, recommendations, and detection of abnormalities. The accessibility, transparency, and, not least, versatility make it a highly essential tool in eclectic work in machine learning and for its application in many other areas. Furthermore, things are moving ahead in the research direction to make k -NN more robust and successful in fields such as data science and artificial intelligence, which are changing fast.

4.3.2. Gradient boosting (GB)

Gradient Boosting Machines (GBM) remain strong machine learning models for handling regression and classification problems. The

performance of GBM models depends much on the fine-tuning of their hyperparameters, which is necessary to strike a balance between biases and variances and, thus, to optimize the model's performance. Key hyperparameters include the number of trees, their depth, the learning rate, and the minimal samples per leaf. Optimal calibration of these parameters can lead to a great deal of the model's accuracy, so systematic analysis is required in order to see the impact of these parameters on performance. The article provides a comprehensive guide on stepwise tuning and evaluating these parameters in Python, with the objective of improving the predictive accuracy of GBM models. It brings out the need to understand the mechanics of the algorithm, the meaning of each parameter, and practical techniques of hyperparameter optimization. This guide is an invaluable tool for data scientists and machine learning enthusiasts keen on utilizing the complete functionality of GBM in their tasks. GBM are built to progressively improve a loss function over the function space by choosing a function referred to as a weak learner that moves in the opposite direction of the gradient. undefined

1. Initialization: Initiate with a baseline model $F_0(x)$, which is typically the target variable's average (for regression tasks) or log odds (for classification tasks).

2. Iterative Process for $m = 1$ to M (M is the total number of iterations for boosting):

Box 2

Output of the Second Weak Learner.

After the first iteration, the weights of the data points are adjusted: points correctly classified previously are assigned a lower weight, and the misclassified points receive a higher weight. The second learner thus prioritizes the higher-weighted points, aiming to classify them correctly. However, this focus shifts away from the points that were correctly classified in the first iteration, potentially leading to their misclassification.

This pattern continues through subsequent iterations (as hinted at by the similar trend in Box 3), with each weak learner focusing on the errors of its predecessors. Each learner is assigned a weight based on its accuracy, with more accurate models receiving higher weights. After many iterations, the outputs of all weak learners are combined, with their respective weights considered, to produce a final, aggregated result. This cumulative approach allows for boosting and effectively enhancing prediction accuracy by concentrating on the most difficult-to-classify points throughout the training process.

The parameters of an ensemble model, particularly in the context of boosting algorithms like GBM or random forests, play a crucial role in defining the model's performance and behavior. These parameters can be broadly categorized into three groups:

1. **Tree-Specific Parameters:** These parameters directly influence the construction and characteristics of each tree within the ensemble. Key tree-specific parameters include:

Maximum depth of the tree that controls the maximum number of levels in each tree, minimum samples split at which the minimum number required to split an internal node, minimum samples leaf at which the minimum number needed to be at a leaf node, and maximum number of features at which the number to consider when looking for the best split.

2. **Boosting Parameters:** These parameters are central to the boosting process and affect how the ensemble of trees is built and combined. Important boosting parameters include:

Learning rate that determines the contribution of each tree to the outcome and controls the rate of correction at each step, often balancing speed and accuracy, number of estimators, which is the total number of trees to be included in the model, subsample at which the fraction of samples to be used for fitting each base learner, if less than 1.0, introduces stochasticity into the model which can help in preventing overfitting.

3. **Miscellaneous Parameters:** These parameters encompass various other aspects of the model's functioning that don't fall strictly under tree construction or the boosting process. This may include:

Loss function that specifies the loss function to be optimized, which can vary based on the type of problem (regression, classification), random state at which the seed of the pseudo-random number generator is used for random sampling, ensuring reproducibility of the model's results, and verbosity that controls the level of messages (such as progress and performance metrics) output by the model during training.

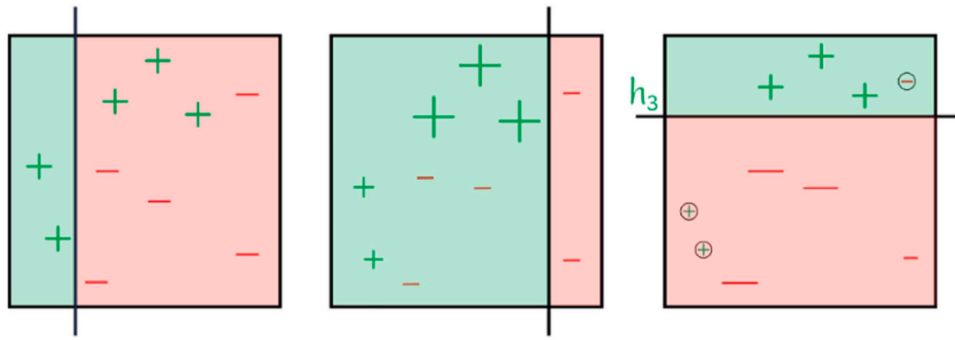


Fig. 3. : GB Idea Visualization.

- Calculate the residuals (r_{im}), defined as the negative gradient of the loss function $L(y_i, F(x_i))$ with respect to $F(x_i)$, evaluated at the model from the previous iteration $F_{m-1}(x_i)$
- Train a weak learner $h_m(x)$, Such as a decision tree on these residuals.
- Enhance the model by incorporating a scaled variant of this weak learner (Bashir et al., 2021)

$$F_m(x) = F_{m-1}(x) + \nu h_m(x) \quad (2)$$

Where ν represents the learning rate.

3. Final Model Output: Produce $F_m(x)$, the aggregation of the initial model and all subsequent scaled weak learners. Through this process, GBM meticulously addresses the inaccuracies of prior iterations, progressively refining the model's precision.

Boosting operates as a sequential method under the ensemble learning framework, aiming to enhance prediction accuracy by amalgamating multiple weak learners. At any given step, t , the predictions from the model are adjusted according to the performance at the previous step, $t-1$. Correctly predicted outcomes receive a reduced weight, whereas misclassified outcomes are assigned a higher weight. This approach is systematically applied to classification problems, with an analogous strategy employed for regression scenarios. Through this iterative weighting and adjustment, boosting effectively focuses the learning process on the more challenging aspects of the training data, progressively improving the overall model's ability to predict accurately. Fig. 3 shows the boosting idea. The process unfolds as follows:

4.3.3. Random forest (RF)

Among the key algorithm types in machine learning, random forest is one of the most famous due to its flexibility and high accuracy. The several different decision trees that it combines make it very good at classifying and handling variety. The flexibility of its application in both classification and regression tasks is widely observed, which promotes the unique feature of the model that can be applied comprehensively in many predictive modeling projects (Mohsen et al., 2023a).

Among the key factors to be considered for the use of the random forest is that the trees have inherent limitations; that is, individual decision trees easily overfit, which, as a result, affects the accuracy. The classifier will make an excessively accurate model. When such an occurrence happens, the training data is idiosyncratic—data that is well-represented by the given model. Random Forest reduces the danger of being detrimentally dependent upon a particular individual decision tree by effectively utilizing the principle of ensemble learning to boost the prediction precision and the reliability of the model.

The main idea of the Random Forest algorithm lies in the principles of ensemble learning, which are a collection of decision trees that have a combined effect on the algorithm's predictive capabilities. Owing to the principle of independence, there is a separate decision tree in the ensemble, which was trained on a randomly chosen subset of the training data and features. This deliberate injection of randomness

ensures that the model trees do not have any extraneous patterns in them, which otherwise could lead to overfitting. The randomness enables the model to have broader applicability by generalizing to unseen data.

Random Forest regression involves accumulating inner tree ensembles' outputs before making predictions. Regression tasks usually involve averaging predictions over the ensemble; that is, the group of learners with contrasting variations will mitigate individual variations to achieve a more stable and accurate prediction. Conversely, for the labeling tasks, the voting strategy is very widely used, where the class label, more commonly among the ensemble votes, is selected as the final output.

The Random Forest is a powerful machine that marries a large number of individual decision trees together to squeeze the most out of these separate branches and get better accuracy in prediction. Due to its versatility across multiple domains, including predictive maintenance for industries and medical diagnostics for the healthcare domain, it has proved to be highly effective in addressing real-world problems. Advancements in machine learning, in turn, keep moving forward. However, Random Forest turns out to be a strong technique that can elegantly cope with a wide range of analysis tasks at these moments just because of its absolute precision and predictability.

4.4. Naive Bayes

A naive Bayes Classifier operates on the basis of Bayes' theorem, a foundational result of Statistics that demonstrates how to condition the probability of a hypothesis based on new evidence (Chebil et al., 2023). In short, Bayes' Formula allows us to find the new probability of the event in question after we learn additional information, such as previous probability, the probability of the observations we have taken, and all the probabilities of these observations. This theorem is considered to be the most fundamental one in the collection of tools. It is applied in the identification tasks where the ultimate accord is a class or a category considered for an input whose likelihood serves as a guide.

The formula, as adapted for use in Naive Bayes classifiers, is typically stated as (Chebil et al., 2023)

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \quad (3)$$

In this formulation:

- $P(A|B)$ is the posterior probability of class A , given predictor B .
- $P(B|A)$ is the likelihood which is the probability of predictor B given class A .
- $P(A)$ is the prior probability of class A .
- $P(B)$ is the prior probability of predictor B .

The Naive Bayes method used for classification is based on the assumption that every feature in the dataset is free from all other fea-

tures given the target class, the class predicted. Although this supposition might come across as being simplistic, it indeed makes many calculations easier and, therefore, reduces computational time even if it is given large datasets to process. The condition for this is the assumption that the characteristics are independent and that the joint probability of a feature set given a class is represented as the product of conditional probabilities of each feature given the class (Chebil et al., 2023)

$$P(X|Y) = P(x_1|Y) \times P(x_2|Y) \times \dots \times P(x_n|Y) \quad (4)$$

Here, X represents the set of features $x_1, x_2, \dots, x_n$, and Y is the class. Each term $P(x_i|Y)$ represents the probability of observing the particular feature x_i in the instances of class Y .

The decision rule for classification using Naive Bayes, then, is straightforward. For an input feature set, the classifier calculates the product of the feature likelihoods and the class prior probability for each class. The class that maximizes this product is chosen as the predicted class (Chebil et al., 2023)

$$\hat{Y} = \arg \max_Y P(Y) \prod_{i=1}^n P(x_i|Y) \quad (5)$$

This product combines:

- $P(Y)$, the prior probability of class Y , reflecting the general prevalence of each class in the training set.
- $\prod_{i=1}^n P(x_i|Y)$, the likelihood of the observed features, calculated as the product of the probabilities of each feature given the class.

Although such a proposition is simplistic, the Naive Bayes model can show unparalleled performance even in many real-life situations. Well-performed AI systems in these tasks are typically related to spam detection as well as document segmenting and result from characteristics with high dimensionality and an abundance of features that make a complex system impracticable. The model can model situations with different characteristics through its use of prediction rankings compared to the system with super exact probability valuations. In this case, the model does not need to predict the exact probabilities but instead the correct relational ones in accordance with classes.

The success of the Naive Bayes model is heavily influenced by the simplicity of its features and the statistical properties among the classes. Even in situations where the features are imperfectly uncorrelated, a common occurrence, the model can still produce remarkably accurate predictions if the predictive power of the features outweighs the inaccuracies introduced by the independence assumption. This adaptability makes Naive Bayes not just a highly applicable model, but also an essential component in the machine learning array. It provides a solid foundation for making comparisons and deriving insights at the outset of the predictive modeling exercise.

4.5. Unconstrained function optimization

Optimization algorithms are essential mathematical instruments of machine learning and data analysis that aim to choose the best solution among a number of opportunities. These tools play a crucial role in unconstrained function optimization and feature selection tasks, where they are used for the maximization or minimization of any function, either without explicit constraints or within an embedded constraint framework. This essay analyses the basics of the optimization algorithms, their applications, and the comparative discussion, emphasizing issues of significance and the challenges of the implementation. This type of optimization is aimed at the extremum of a function that is not determined by any restrictions on the values of the variables. It is a fundamental issue in all the scientific and engineering areas focused on the optimal results of the parameter variations. Gradient Descent is noted for its simplicity and efficiency. It walks towards the global minimum of the function in a direction opposite to the gradient. Its

alternatives, like the stochastic gradient descent and Mini-batch gradient descent, increase convergence and computational efficiency, making it irreplaceable in large-scale machine learning tasks.

Newton's Method improves the convergence rate for the function by taking into account the second-order Taylor series expansion. Still, it is limited by the computation load of the Hessian matrix for high dimensions. Quasi-Newton Methods, such as Broyden Fletcher Goldfarb Shanno (BFGS), approximate this matrix and so essentially compromise between computational speed and accuracy. Suitable for quadratic forms but more general, the conjugated gradient method is a highly successful large-scale optimization method that updates solutions in conjugate directions to minimize the iteration number for convergence. Choosing the appropriate features is very important to improve the performance of the model and to reduce its complexity and computational requirements. Filter methods determine the relevance of features by their intrinsic attributes, employing such metrics as correlation coefficients and mutual information. They are computationally efficient but do not always represent feature interactions. However, while wrapper methods such as forward selection and recursive feature elimination do provide a more detailed approach by taking interactions into account, they are more computationally intensive.

Most of the mentioned methods that integrate feature selection into the training process, such as the Lasso regression and Decision Trees, naturally ignore non-relevant features and thus represent a compromise between the efficiency of filter methods and the effectiveness of wrapper methods. The appropriateness of optimization algorithms and feature selection techniques depends on the type of problem, as well as the characteristics of data and available computational resources. Even though many gradient descent variants are generally applicable, the y may fail for non-convex functions or illusive gradients. Newton's Method and its quasi-Newton modifications are faster in convergence but are computationally expensive. Approaches to feature selection are problem-dependent, combining performance and computational practicality. Implementational obstacles consist of hyperparameter tuning, steering non-convex optimization landscapes, and avoiding overfitting in feature selection.

5. Simulation results

Grid Simulated Dataset, many classifiers have been used to evaluate the effectiveness, whereas the performance of these algorithms is assessed using the train-test split methodology. The dataset was split into two segments, with 30 % set aside for testing and the rest of the 70 % for training. The first subset selected for training helps in assessing the appropriateness of the model using the input features of the dataset. Conversely, the second subset, intended for simulating scenarios, is devoid of the dataset's input component, facilitating a comparison between actual outcomes and the model's predictions. This subset represents the test dataset. Additionally, the efficiency of the gradient boosting technique has been scrutinized alongside other algorithms, with the SG data being partitioned into an 80:20 split for training and testing purposes, respectively. Several experiments have been undertaken to showcase the effectiveness and superior performance of the recommended algorithm. The trials were executed on a machine equipped with Windows 10 and Python 3.9, powered by an Intel® Core™ i7 CPU operating at 3.00 GHz. This hardware is manufactured by Intel Corporation, located in California, United States. Model evaluation is a crucial process in any computational task, particularly predictive ensemble modeling. This type of process, when the emphasis is on the analysis of variability and various performance measures, is especially critical. The evaluation metrics are based on four fundamental categories: TP , TN , FP , and FN . Confusion matrix is a popular tool for performance measurement of the given model with the help of accuracy (Mohsen et al., 2023b; Parmar et al., 2019).

Table 1

Evaluation of the proposed classification method using the SG dataset.

Models	Accuracy	Sensitivity (TRP)	Specificity (TNP)	P-value (PPV)	N-value (NPV)	F-score
Logistic Regression	0.79787234	0.885093168	0.608108108	0.83090379	0.708661417	0.857142857
Naive Bayes	0.811088296	0.897058824	0.612244898	0.842541436	0.72	0.868945869
AdaBoost	0.821705426	0.904891304	0.614864865	0.853846154	0.722222222	0.878627968
Support Vector Machine	0.830508475	0.907894737	0.635761589	0.8625	0.732824427	0.884615385
K-Nearest Neighbors	0.861204013	0.922897196	0.705882353	0.887640449	0.784313725	0.904925544
Decision Tree	0.880126183	0.932017544	0.747191011	0.904255319	0.81097561	0.917926566
Neural Network (MLP)	0.892957746	0.941729323	0.747191011	0.917582418	0.81097561	0.929499072
Random Forest	0.910757946	0.955414013	0.763157895	0.930232558	0.838150289	0.942655145
Gradient Boosting	0.924369748	0.958466454	0.821256039	0.941915228	0.867346939	0.950118765

- Accuracy: Evaluates how well the model can recognize full cases, irrespective of whether they are positive or negative and may be represented as (Zaki et al., 2023)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

- Sensitivity: It is often referred to as the "true positive rate" (TPR). The capability of the positive case is computed by (Zaki et al., 2023)

$$Sensitivity = \frac{TP}{TP + FN} \quad (7)$$

- Specificity: This is the real negative rate, otherwise known as selectivity. It gains the capacity to locate cases in which there is no match. It is calculated as follows (Zaki et al., 2023):

$$Specificity = \frac{TN}{TN + FP} \quad (8)$$

- Precision: It is the value of making correct predictions. It controls the proportion of genuine positives relative to the total number of positive results. It is computed as follows (Zaki et al., 2023):

$$PPV = \frac{TP}{TP + FP} \quad (9)$$

- Negative Predictive Value (NPV): It controls the proportion of real negatives relative to other negative values (Zaki et al., 2023).

$$NPV = \frac{TN}{TN + FN} \quad (10)$$

- F-score: It calculates the average harmonic value of the precision and sensitivity measurements (Zaki et al., 2023).

$$F - score = 2 \frac{PPV \times TPR}{PPV + TPR} \quad (11)$$

After conducting a thorough comparative analysis between the

outcomes obtained from our proposed algorithm and those documented in the preceding study, the comprehensive examination revealed not only quantitative advantages but also qualitative enhancements, thereby solidifying the efficiency and reliability of our proposed approach. These findings signify a notable advancement in the field, underscoring the significance and potential impact of the novel methodology I have introduced. As depicted in Table 1, a comparison of different machine learning models is quite clear due to their performance metrics on different aspects of the SG dataset. This table compares the proposed classifying technique by showing the main statistics, which are sensitivity (True Positive rate, TPR), specificity (True Negative rate, TNP), positive and negative predictive values, and the F-score for every type of model. The models to be checked consist of Logistic Regression, Naive Bayes, AdaBoost, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Neural Network, i.e. MLP, Random Forest, and Gradient Boosting.

A thorough analysis of the performance metrics for predictive models reveals that Gradient Boosting stands out with an accuracy of 92.44 %, surpassing the next best model, Random Forest, by approximately 1.4 %, with an accuracy of 91.08 %. The significantly lower accuracy of Logistic Regression at 79.79 % indicates it might be less suited for this specific dataset. Gradient Boosting shines in sensitivity, achieving a remarkable 95.85 %, which is around 7 % higher than that of Logistic Regression, highlighting its efficiency in identifying true positives—a crucial aspect in situations where missing true positives could have significant consequences. On the specificity front, which measures the true negative rate, Gradient Boosting's performance at 82.13 % is notably superior to Logistic Regression's 60.81 % by more than 20 %, showcasing its capability in accurately classifying negative instances in imbalanced datasets. When evaluation of Positive Predictive Value (PPV) and Negative Predictive Value (NPV) is done, Gradient Boosting has the highest PPV of 94.19 % among such metrics, which also shows that the rate of false positives in this model is lower than that in Naïve Bayes which has PPV of 84.25 %. Its NPV at 86.73 % also shows a much stronger predicting capability than Logistic Regression's 70.87 %, taking advantage of more accurate negative predictions. Next, with the highest F-score value of 95.01 % against Logistic Regression's 85.71 % for this dataset, Gradient Boosting shows a good compromise between precision and recall, thus proving to be the best option for managing both types of errors for this sample.

Table 2

Evaluation of the proposed hyperparameter optimization method.

Models	Accuracy	Sensitivity (TRP)	Specificity (TNP)	P-value (PPV)	Nvalue (NPV)	F-score
BA+GB	0.930588235	0.961904762	0.840909091	0.945397816	0.885167464	0.953579858
WAO+GB	0.935747664	0.964968153	0.855263158	0.948356808	0.898617512	0.956590371
BBO+GB	0.941109853	0.970724191	0.858974359	0.950226244	0.913636364	0.960365854
MVO+GB	0.947426068	0.973053892	0.87755102	0.955882353	0.922746781	0.964391691
SBO+GB	0.951612903	0.975975976	0.890151515	0.957290133	0.93625498	0.966542751
GA+GB	0.957011259	0.978417266	0.904255319	0.961810467	0.944444444	0.970042796
PSO+GB	0.963451777	0.981240981	0.921232877	0.967283073	0.953900709	0.974212034
GWO+GB	0.971028037	0.981919332	0.948717949	0.975138122	0.962427746	0.978516979
DTO+GB	0.993171943	0.991649269	0.995405819	0.996852046	0.987841945	0.994243851

Table 3
Statistical analysis of the proposed hyperparameter optimization method.

	BA+ GB	WAO+ GB	BBO+ GB	MVO+ GB	SBO+ GB	GA+ GB	PSO+ GB	GWO+ GB	DTO+ GB
Number of values	10	10	10	10	10	10	10	10	10
Minimum	0.9291	0.9306	0.9391	0.9404	0.9492	0.9507	0.9593	0.9681	0.9912
25 % Percentile	0.9306	0.9344	0.9411	0.9474	0.9516	0.957	0.9603	0.971	0.9932
Median	0.9306	0.9357	0.9411	0.9474	0.9516	0.957	0.9635	0.971	0.9932
75 % Percentile	0.9306	0.9367	0.9424	0.948	0.952	0.9573	0.9635	0.971	0.9932
Maximum	0.9366	0.9426	0.9461	0.9507	0.9582	0.9637	0.9635	0.9831	0.9952
Range	0.007529	0.012	0.007	0.01032	0.009	0.013	0.004107	0.015	0.004
10 % Percentile	0.9292	0.9309	0.9393	0.9411	0.9494	0.9513	0.9594	0.9684	0.9914
90 % Percentile	0.936	0.9423	0.9461	0.9506	0.9577	0.9631	0.9635	0.9819	0.995
Actual confidence level	97.85 %	97.85 %	97.85 %	97.85 %	97.85 %	97.85 %	97.85 %	97.85 %	97.85 %
Lower confidence limit	0.9306	0.9337	0.9411	0.9474	0.9516	0.957	0.9603	0.971	0.9932
Upper confidence limit	0.9306	0.9396	0.9461	0.9497	0.9532	0.958	0.9635	0.971	0.9932
Mean	0.931	0.936	0.9419	0.9473	0.9522	0.9571	0.9624	0.9719	0.9932
Std. Deviation	0.002009	0.00321	0.0023	0.00269	0.00231	0.00308	0.00168	0.00402	0.000943
Std. Error of Mean	0.000635	0.00101	0.000727	0.00085	0.00073	0.00097	0.00053	0.00127	0.000298
Lower 95 % CI of mean	0.9296	0.9337	0.9403	0.9454	0.9505	0.9549	0.9612	0.9691	0.9925
Upper 95 % CI of mean	0.9325	0.9383	0.9436	0.9492	0.9538	0.9594	0.9636	0.9748	0.9938
Geometric mean	0.931	0.936	0.9419	0.9473	0.9522	0.9571	0.9624	0.9719	0.9932
Skewness	2.787	0.6779	1.401	-1.905	2.096	0.06571	-1.141	2.821	-2.2E-13
Kurtosis	8.598	1.692	1.019	5.637	6.101	4.323	-0.6913	8.712	4.5
Sum	9.31	9.36	9.419	9.473	9.522	9.571	9.624	9.719	9.932

Gradient boosting does better than the competitors in the broad nature of benchmarks. It has the highest accuracy at 92.44 %, which corresponds to the highest level that is necessary for its overall predicting quality. Its superiority is highlighted by leading the field in this contribution in terms of sensitivity (95.85 %) and specificity (82.13 %), which testifies to the incredible nature of the trichotomous model in differentiating between positive and negative cases. The model also has a great scene of 95.01 %, which shows the model’s ability to provide a perfect balance between precision and recall. The convincing metrics provide us confidence that Gradient Boosting should then be selected as our model of choice. Hyperparameter optimization techniques will also be another means of going further with our capacities. This will permit the processing of the model’s parameters for fine-tuning to improve the model’s predictability and achieve the best results aimed at our dataset and objectives. Table 2 gives different solutions for the problem with GB and some optimization techniques.

Hyperparameter optimization techniques with the Gradient Boosting model have proven to be really effective in improving the model’s performance metrics. Out of all the tested models, the DTO+GB model clearly stands out with the scores. It shows a human accuracy of 99.32 %, a sensitivity rate of 99.16 %, and an F-score of 99.42 %, which are amazing scores. This implies that the DTO optimization algorithm is indeed better in terms of striking a balance between the optimality of the fit and the model itself. The same goes for the GWO+GB, which also gives outstanding results; its accuracy is 97.10 %, and specificity is 94.87 %, which shows the high effectiveness of the GWO hyperparameter optimization. The improvements noticed for all models additionally bring into sharp focus the significant effect of hyperparameter tuning in taking the baseline performance of the gradient-boosting model to its highest levels.

Moreover, the DTO+GB model post-optimization chest shows tremendous improvement in its PPV and NPV parameters. The PPV reaches as high as 99.68 %, and the NPV is around 98.78 %. Hence, this points out the need for the incorporation of hyperparameter optimization as part of a model since DTO and GWO are the most efficient in achieving the goal of accuracy and dependability with this specific data. Statistical tests serve to validate the efficiency of the feature selection approach, which is introduced, and it shows the importance of supporting this feature in improving the model’s prediction capabilities. The statistically quantifiable figure received from both the tests and the adopted methodology re-affirming that the feature selection is working its magic, which in turn is boosting the performance of models aimed at the detection of SG stability, as portrayed in Table 3.

Table 4
ANOVA of The Proposed Feature Selection Method Based on the SG dataset.

	SS	DF	MS	F (DFn, DFd)	P value
Treatment (between columns)	0.02978	8	0.003723	F (8, 81) = 543.4	P<0.0001
Residual (within columns)	0.000555	81	0.000006852		
Total	0.03034	89			

Merging of choice of ANOVA and use of Wilcoxon signed-rank tests helps in analyzing different statistical points and the significance of the proposed feature selection methodology. Generally speaking, the analyses are structured around two contrasting hypotheses that are referred to as H0 (the null hypothesis) and H1 (the alternative hypothesis). The results of the tests (Tables 4 and 5), shown below, allow the generation of statistics that demonstrate the empirical significance of the suggested method for choosing the set of features important for a SG stability classification.

The tables illustrate the results of this feature selection approach, which presents reliability in the success of the SG stability prediction models based on enhanced performance indicators. The statistical method ANOVA, which is good for comparing the means across many groups, and the Wilcoxon signed rank test, which computes differences between paired observations, are used in the analysis, which provides robust proof to the hypothesis that the chosen feature selection process contributes perfectly to the improvement of classification accuracy and reliability.

The data presented thoroughly evaluates the gradient-boosting performance indicators for models optimized around hyperparameters covering a scale from BA+GB to DTO+GB. A ten-value metric performance for each model allows for a reliable foundation for comparison. The worth of the DTO+GB model lies in the fact that its median performance value, which stands at 0.9932, has the same value in its 25th and 75th percentiles, which implies that its performance is consistently stable. It further bests the record of the highest maximum value of 0.9952 and a minimal range of 0.004, thereby vouching for its excellence and excellence.

Standard deviation measures across the models reveal that the DTO+GB model exhibits the smallest variation around the mean (0.000943), underscored by the lowest standard error of the mean (0.000298). This indicates a high level of precision in estimating the true

Table 5
Wilcoxon of the proposed feature selection method based on the EV population dataset.

	BA+GB	WAO+GB	BBO+GB	MVO+GB	SBO+GB	GA+GB	PSO+GB	GWO+GB	DTO+GB
Theoretical median	0	0	0	0	0	0	0	0	0
Actual median	0.9306	0.9357	0.9411	0.9474	0.9516	0.957	0.9635	0.971	0.9932
Number of values	10	10	10	10	10	10	10	10	10
Sum of signed ranks (W)	55	55	55	55	55	55	55	55	55
Sum of positive ranks	55	55	55	55	55	55	55	55	55
Sum of negative ranks	0	0	0	0	0	0	0	0	0
P value (two-tailed)	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
Exact or estimate?	Exact	Exact	Exact	Exact	Exact	Exact	Exact	Exact	Exact
Significant (alpha=0.05)?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Discrepancy	0.9306	0.9357	0.9411	0.9474	0.9516	0.957	0.9635	0.971	0.9932

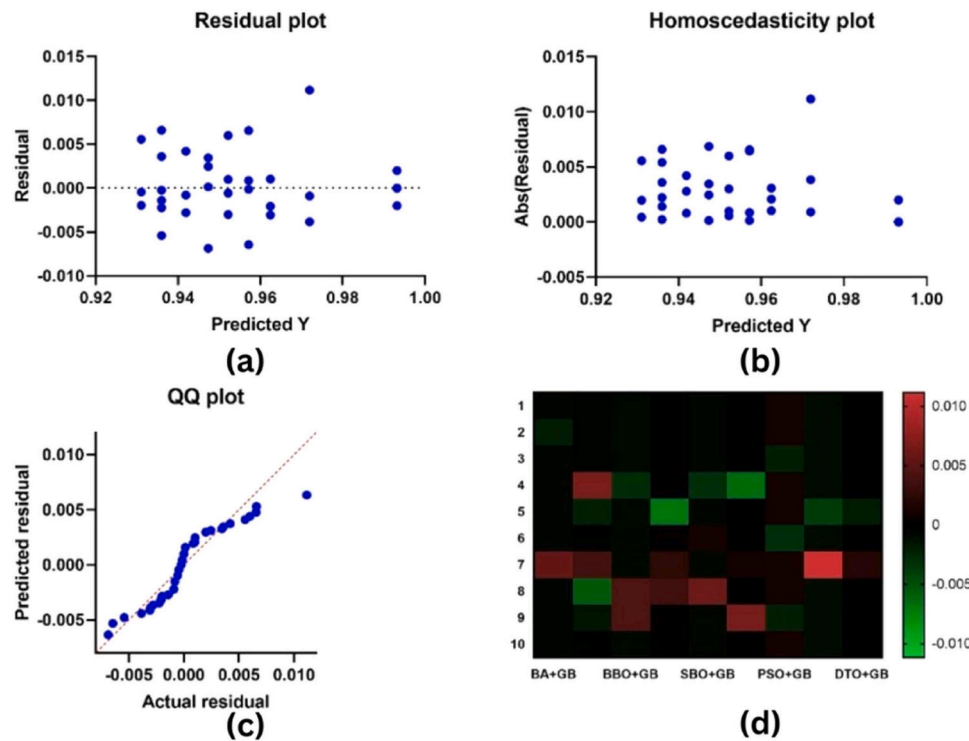


Fig. 4. : Visualizing The Performance of The Proposed Feature Selection Method Applied to The SG Dataset.

mean performance. The confidence intervals for DTO+GB are notably tight, suggesting a symmetric data distribution without significant skewness or heavy tails. The skewness and kurtosis metrics also vary across the models, with DTO+GB showing a skewness near zero, which points to a symmetrical distribution around the mean. Its kurtosis value of 4.5 suggests a balanced distribution, neither excessively peaked nor flat, indicative of a well-behaved data set relative to a normal distribution. Table 4 gives the ANOVA test results.

The ANOVA analysis reveals a highly significant impact of the treatments on the outcomes measured, indicated by a Treatment Sum of Squares (SS) of 0.02978 over 8 degrees of freedom (DF) and a resulting Mean Square (MS) of 0.003723. This leads to an F-ratio of 543.4, calculated for 8 and 81 degrees of freedom for the numerator and denominator, respectively. This pronounced F-value, coupled with a P-value of less than 0.0001, decisively refutes the null hypothesis, indicating that the observed differences in treatment effects are statistically unlikely to be attributable to random variation. The notable gap between the treatment mean square and the residual mean square, which stands at 0.000006852, further validates the strength and statistical significance of the treatment effect. These compelling statistics provide robust evidence that the treatments exert a significant and measurable impact on the observed results.

The data indicates that implementing various hyperparameter optimization strategies on a gradient-boosting model substantially enhances its performance. These strategies have elevated the model’s median performance metrics to impressive levels, ranging from 0.9306 to 0.9932, well above the baseline theoretical median of 0. The application of the Wilcoxon Signed Rank Test, yielding a uniform sum of signed ranks (W) of 55 for each model and demonstrating no negative ranks alongside a P value of 0.002—significantly lower than the conventional alpha threshold of 0.05—confirms the optimizations’ effectiveness. This statistical outcome firmly establishes that the performance medians for these models are considerably higher than zero, confirming the optimizations’ success.

Fig. 4-a shows the Quantile-Quantile (QQ) Plot, which highlights slight deviations from the ideal 45-degree line at the extreme quantiles. This indicates that the residuals’ distribution has heavier tails than a normal distribution. This deviation is particularly marked at the upper quantiles, where residuals depart from the expected line, suggesting a higher occurrence of extreme values than would be typically expected in a normally distributed dataset.

The scatter plot 4-b offers a detailed visual analysis comparing the accuracy of various hyperparameter-optimized gradient-boosting algorithms. Accuracy, likely the metric of interest, is plotted on the y-axis,

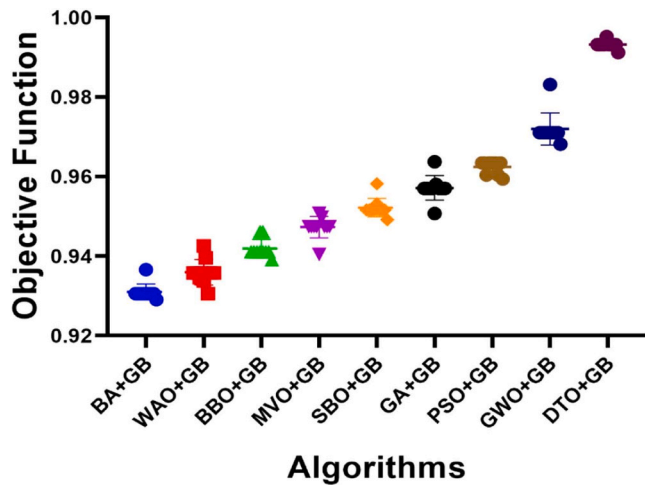


Fig. 5. : Visualizing The Accuracy Values of The Proposed DTO-GB Optimization Algorithm.

allowing for a direct comparison of each algorithm's performance. The BA+GB algorithm, represented by blue circles in Fig. 4-c, shows accuracy levels ranging from around 0.92 to just over 0.94. This range indicates variability in performance across different iterations. WAO+GB, shown with red squares, presents a more concentrated cluster of results, with accuracy levels slightly higher, mostly between 0.94 and 0.95. The BBO+GB algorithm, depicted with green triangles, also falls within the 0.94–0.95 accuracy range but displays a broader distribution that points to variability in its performance from one run to another.

In Fig. 4-d, algorithms with generally higher accuracy rates are revealed. GA+GB, marked by black circles, delivers consistent accuracy around the 0.96 level, showcasing reliable performance. The PSO+GB and GWO+GB algorithms, indicated by brown crosses and navy horizontal ellipses, respectively, impress with a tight grouping of data points around the 0.97 accuracy level. This suggests both high performance and minimal variation across runs. Most notably, the DTO+GB algorithm, visualized with purple clouds, stands out significantly from the others, with accuracy points densely congregated just below the 1.00 mark. This exceptional performance indicates not only the highest accuracy among the compared algorithms but also remarkable stability in its results. The distribution of data points for each algorithm highlights the performance variability inherent to each method. The least variation and highest performance consistency are observed in the PSO+GB, GWO+GB, and especially the DTO+GB algorithms, identifying

DTO+GB as the top performer in this comparison. This visualization effectively underscores the comparative effectiveness and reliability of these hyperparameter-optimized gradient-boosting algorithms in achieving high accuracy rates.

Fig. 5 shows the histogram that summarizes the accuracy distributions for various hyperparameter-optimized gradient boosting, with distinct colors representing each optimization algorithm. The x-axis outlines the accuracy scores from roughly 0.928–0.996, while the y-axis indicates the count of occurrences within each accuracy bin. The BA+GB algorithm, depicted in blue, demonstrates a range of values, predominantly peaking around the 0.930–0.932 accuracy segment. Similarly, the WAO+GB algorithm, marked in red, exhibits a concentration of values primarily around the 0.932–0.934 accuracy range. Both BBO+GB, shown in green, and MVO+GB, in purple, reveal broader distributions, with the most frequent values for BBO+GB at 0.938–0.940 and for MVO+GB at 0.946–0.948. The SBO+GB algorithm, presented in orange, though less populated, registers consistently high values, peaking at the 0.950–0.952 accuracy bin. At the upper end of the accuracy spectrum, the algorithms GA+GB (black), PSO+GB (brown), GWO+GB (navy), and particularly DTO+GB (purple) show more sparse distributions but reach higher accuracy levels. GA+GB clusters around the 0.956–0.958 range, indicating tight grouping at this accuracy level. PSO+GB, although with fewer instances, falls within the higher accuracy bracket of 0.968–0.970. Fig. 6

This histogram vividly delineates each hyperparameter-optimized GB algorithm's variability and central accuracy tendencies. Among them, DTO+GB emerges as the standout performer, achieving the highest and most focused accuracy scores, underscoring its exceptional effectiveness relative to the other algorithms.

Fig. 7, a pair plot with corresponding regression lines, compares the DTO-GB optimizer performance with those of other competitive algorithms. Pairplot visualizations can be very useful for researchers who need to study correlations between many variables in a dataset. In this situation, each plotting point depicts something about a single optimization algorithm's performance metrics. The visualization of the regression lines encourages the study of the patterns and the interactions among these algorithms. By performing this visual comparison, researchers can evaluate how the DTO-GB approach stands against other methods for the best optimization performance and pinpoint any structural patterns or outliers that can factor in effectiveness.

Fig. 8 shows a Kernel Density Estimation (KDE) plot demonstrating the probability distribution of accuracy scores of the proposed DTO-GB optimization and other competing algorithms. KDE plots offer a view of the probability density function of the continuous variable being studied - in this scenario, accuracy scores. Researchers can apply the violation

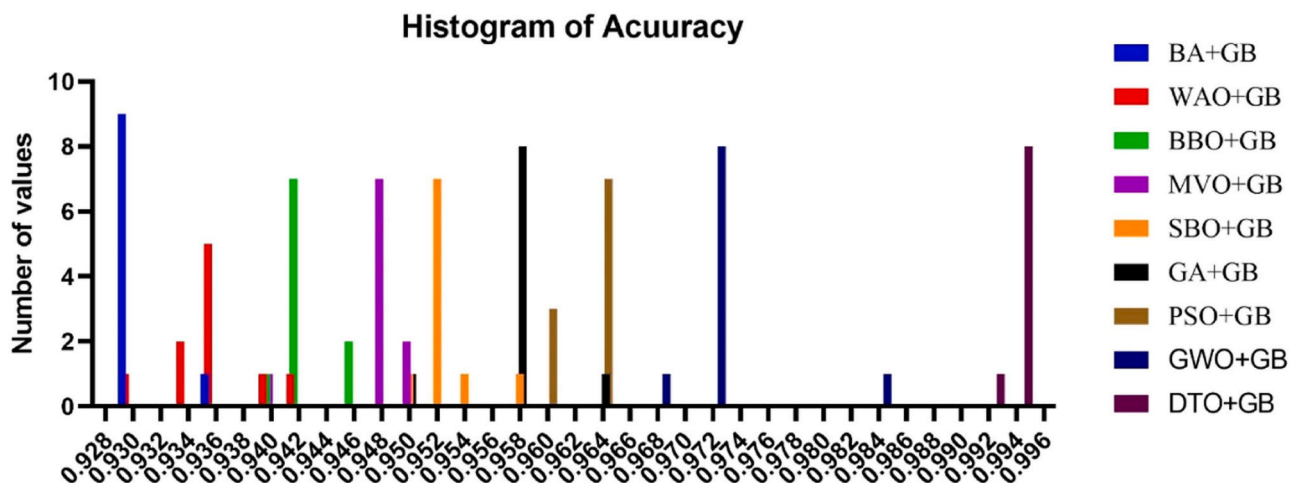


Fig. 6. : Histogram of The Accuracy Values for The Proposed DTO-GB Optimization Algorithm and other Competitive Ones.

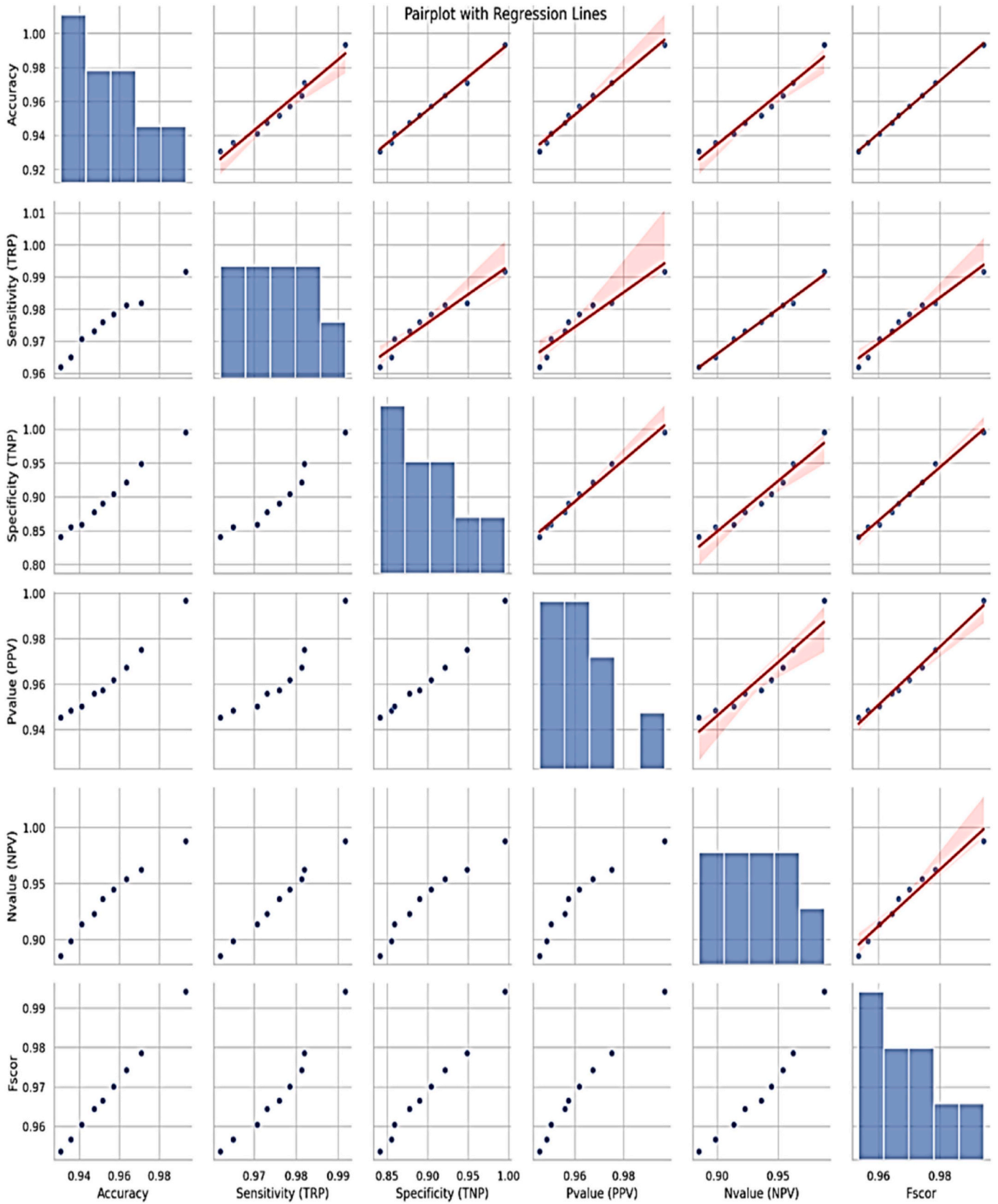


Fig. 7. : Pairplot with Regression Lines for the Proposed DTO-GB Optimization Algorithm and other Competitive Ones.

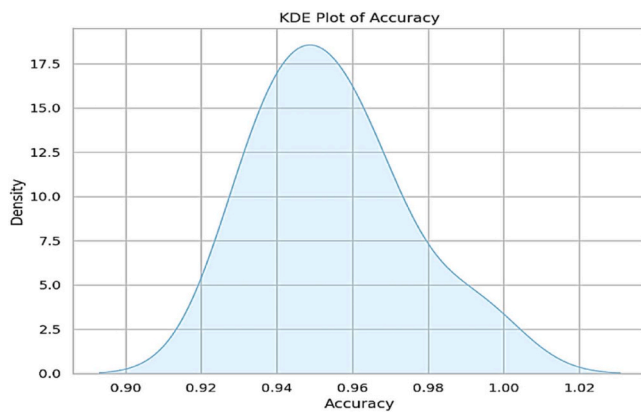


Fig. 8. : KDE Plot of Accuracy for The Proposed DTO-GB Optimization Algorithm and Other Competitive Ones.

technique to uncover central tendencies, variabilities, and relative performance levels among the algorithms through the visualization of the distributional characteristics of accuracy scores. It enables informed choice about algorithms, creating the possibility of developing optimization techniques. Because of this, optimization advances.

This research provided affirmative evidence that highlighted the efficacy of Gradient Boosting models, which are mostly enhanced by Dipper Throated Optimization (DTO) and other hyperparameter tuning methods in forecasting SG stability. Through the DTO+GB model, outstanding performance was attained, which is evident from its accuracy, recall, and F-score measures. The results provide evidence that smartly designed machine learning algorithms have the power to revolutionize the way a SG network operates, with the promise of bringing qualitative results in this area.

6. Conclusions and future work

This research paper carries out a comprehensive analysis of the most promising gradient-boosting hyperparameter-optimized models for predicting SG stability. The DTO+GB model was found to be the best-performing model among all the models that were tested, achieving an accuracy of 99.32 %, sensitivity of 99.16 %, and specificity of 99.54 %. These results provide evidence that smartly designed machine learning algorithms have the power to revolutionize the way a SG network operates, with the promise of bringing qualitative results in this area. Accuracy, sensitivity, specificity, positive predictive value, and negative predictive value were used to show that this model is superior in all the key metrics. Image plots indicate that the best diagnostic output is the one that the DTO+GB model has achieved due to its high precision and high recall, seen to be around 100 % F-Score. Correlation functions were used in the regression analysis, demonstrating the existence of a straight line between sensitivity or specificity and the F-Score, showing the efficiency in the classifier selecting more true positives and negatives, hence its accuracy and predictive power. The Kernel Density Estimate (KDE) Plot of Accuracy, yet again, demonstrates the stable and trustworthy class setup of the DTO+GB model, where most of the high-accuracy scores are in the mainline. The observations that follow show the high efficiency of hyperparameter optimization, batch learning, pre-training, and joint training algorithms specifically applied in the DTO+GB model. These results then augur for the use of AI-based machine learning schemes in SG stability projects where precise and dependable predictive models with optimal value can be obtained through efficient optimization methods. The research not only becomes a basis for the improvement of smart network systems but also represents a model to be followed in future studies where machine learning is an essential tool to solve complex problems in the real world. Please find in the next paragraphs an explanation of research that the future is less

likely to be affected by changing weather patterns, grid topologies, and other variables. The model can become accurate and more stable if the dataset is extended. On the other hand, hybrid models, such as extending gradient boosting with other machine learning algorithms that combine them, might achieve the best results. Exploring real-time settings or geographically different settings for dynamic hyperparameter tuning and testing the models in multivariate applications is also recommended to guarantee the accuracy and scalability of the model. Future work will not be limited to the stability constants of the SG systems, the critical issue of power shedding schemes, and the additional experiments on its final scalability.

CRediT authorship contribution statement

Reem Ibrahim Alkanhel: Writing – original draft, Visualization. **Marwa M. Eid:** Writing – original draft, Methodology, Investigation. **Laith Abualqah:** Writing – review & editing, Validation. **El-Sayed M. El-kenawy:** Writing – review & editing, Project administration, Methodology, Conceptualization. **Mohammed A. Saeed:** Writing – original draft, Software, Formal analysis.

Declaration of Competing Interest

No conflict of interest exists.

Data availability

Data are in a repository as public data at <http://www.kaggle.com/datasets/pcbreviglieri/smart-grid-stability>

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Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R323), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

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